

ichosynth — Development Environment

Complete, up-to-date guide for Windows and macOS

Everything you need to compile, modify, and upload the firmware: from the IDE to the libraries, all the way to your first upload onto the Teensy 4.1.

In a hurry? There's a **script that does everything for you** (Teensy core + libraries + patch + compile-check). Jump to [chapter 3](#). The manual guide is in the chapters that follow.

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1 · What we install

Component	Version	What it's for	Required
Arduino IDE	2.3.x	writing/compiling the firmware	
Teensy support (Teensyduino)	1.61	adds the Teensy boards to the IDE (+ Teensy Loader)	
Libraries	see ch. 6	FastLED 3.9.10 , TeensyPolyphony, etc.	
ResamplingReader.h (patch)	from the repo	avoids nullptr crashes during playback	
Python	3.12 (or 3.13+ with audioop-lts)	wavmaker (sample conversion)	optional
Git / GitHub CLI	latest	cloning and version-controlling the project	optional

ATTENZIONE

Two pitfalls that the guide (and the script) handle for you: 1. **FastLED must be 3.9.10** — the 3.10.x releases break on the Teensy's WS2812Serial path. 2. The two **newdigate** libraries (teensy-variable-playback, teensy-polyphony) must be taken from **GitHub** (same HEAD): the copies in the Arduino registry are out of sync with each other.

2 · Requirements

- **Windows:** Windows 10 or 11, 64-bit. Do **not** use Arduino IDE from the **Microsoft Store**: it's incompatible with Teensy. Use the `.exe/.msi` installer (or winget).
- **macOS:** macOS 10.15 or later; native support for **Intel** and **Apple Silicon** (M1/M2/M3/M4). On macOS, **only** Arduino IDE 2.x is supported.
- **A micro-USB data cable** (not charge-only) for the Teensy 4.1.
- ~1 GB of disk space for the IDE + toolchain + libraries.

3 · The fast track: one script does it all

In the repo, under `scripts/`, there are two scripts that install the Teensy core + all the libraries with the **correct versions**, apply the `ResamplingReader.h` patch, and run the **compile-check**.

First install [arduino-cli](#), then run the script:

Windows (PowerShell):

```
winget install ArduinoSA.CLI
powershell -ExecutionPolicy Bypass -File scripts\setup-dev-env.ps1
```

macOS (Terminal):

```
brew install arduino-cli
chmod +x scripts/setup-dev-env.sh
./scripts/setup-dev-env.sh
```

TIP

If the script completes successfully, skip to [chapter 8.4](#) for the upload. If you'd rather understand each step (or the script fails), continue with the manual guide below.

4 · Installing Arduino IDE

Windows

winget (recommended): `winget install --id ArduinoSA.IDE --source winget` or download the **Windows (.exe/.msi)** installer from <https://www.arduino.cc/en/software>.

ATTENZIONE

Mistake #1: **no Microsoft Store**. That version prevents the Teensy Loader from working.

macOS

Homebrew: `brew install --cask arduino-ide` or download the **.dmg** (universal Intel/Apple Silicon) and drag the app into Applications.

ATTENZIONE

Gatekeeper: on first launch use **right-click -> Open** (once), or Settings -> Privacy & Security -> "Open Anyway".

5 · Adding Teensy support

1. Open Arduino IDE -> **File -> Preferences** (Windows, Ctrl+,) or **Arduino IDE -> Settings** (macOS, Cmd+,).
2. In **"Additional boards manager URLs"** paste:

```
https://www.pjrc.com/teensy/package_teensy_index.json
```

3. Open the **Boards Manager** (icon on the left), search for **"teensy"** and install **"Teensy (for Arduino IDE 2.x.x)"** (1.61). The **Teensy Loader** will appear as well.

NOTA

Drivers: on Windows 10/11 and on macOS **no additional drivers are needed**.

Check: is **Tools -> Board -> Teensy -> Teensy 4.1** present? You're ready.

6 · The libraries

Open the **Library Manager** (Ctrl/Cmd+Shift+I) and install:

Search for	Notes
FastLED	exactly version 3.9.10 (not the latest)
Mapf	float mapping
Switch (by Albert van Dalen)	button/gesture handling (avdweb_Switch)
<i>(only if you use the OLED)</i> Adafruit SSD1306 + Adafruit GFX	optional screen

The two **newdigate** libraries are not installed from the Library Manager (the copies are out of sync): they must be installed from GitHub. The easiest way is the **script** in [chapter 3](#); alternatively, from arduino-cli:

```
arduino-cli config set library.enable_unsafe_install true
arduino-cli lib install --git-url https://github.com/newdigate/teensy-variable-playback.git
arduino-cli lib install --git-url https://github.com/newdigate/teensy-polyphony.git
```

NOTA

Already included in the Teensy core (don't install them separately): Audio, Encoder, WS2812Serial, Wire, EEPROM, SD.

7 · The mandatory step: ResamplingReader.h

The teensy-variable-playback library must be **patched** with the file provided in the repo ([DOCS/ResamplingReader.h](#)), which prevents null-pointer crashes.

Replace the file at:

- **Windows:** Documents\Arduino\libraries\teensy-variable-playback\src\ResamplingReader.h
- **macOS:** ~/Documents/Arduino/libraries/teensy-variable-playback/src/ResamplingReader.h

ATTENZIONE

This must be redone **every time you update** that library. (The script in ch. 3 does it for you.)

8 · Downloading the project and compiling

8.1 Download

```
git clone https://github.com/luigismith/ichosynth.git
```

or **Code -> Download ZIP** from the GitHub page. Open `soundpauli_ni404.ino` with Arduino IDE.

8.2 Compile settings (Tools menu)

Item	Value
Board	Teensy 4.1
USB Type	Serial + MIDIx16
CPU Speed	600 MHz (default)
Port	the Teensy's port (after connecting it)

8.3 (3-encoder build) config.h

In [config.h](#) the 3-encoder build is **already set** (`HAS_ENCODER4 0`); leave it as is. Optional fork features:

```
#define OLED_ENABLED 1           // OLED status screen
#define MIDI_CLOCK_OUT_ENABLED 1 // MIDI clock out (master sync)
```

8.4 Compile and upload

1. Connect the Teensy 4.1 via USB (a **data** cable).
2. **Verify** () to compile, then **Upload** (->).
3. The **Teensy Loader** opens: it usually programs on its own; if it stays waiting, press the

little white button on the Teensy.

4. When the flash finishes, the animation starts on the LED matrix.

TIP

The first compilation is slow (building the core); subsequent ones are fast.

9 · Optional tools (Python, Git)

9.1 Python — for the wavmaker sample converter

In the repo, under `_SDCARD/`, you'll find `wavmaker.exe` (Windows GUI, **no Python needed**) and the `wavmaker_gui.py` / `wavmaker.py` sources.

ATTENZIONE

The Python script uses the `audioop` module, **removed in Python 3.13+**. Solutions: **(a)** use **Python 3.12**, or **(b)** `pip install audioop-lts`.

Windows: `winget install Python.Python.3.12` · **macOS:** `brew install python@3.12`

9.2 Git and GitHub CLI

Windows: `winget install Git.Git (+ GitHub.cli optional)` **macOS:** `xcode-select --install` (includes git) · `brew install gh` (optional)

10 · Checklist and common problems

	Check
	Arduino IDE >= 2.3.x starts up
	Tools -> Board -> Teensy -> Teensy 4.1 present
	The installed FastLED is 3.9.10
	The two newdigate libraries installed from GitHub
	ResamplingReader.h replaced
	USB Type = Serial + MIDIx16
	soundpauli_ni404.ino compiles without errors

Symptom	Cause / fix
Teensy doesn't appear in Boards Manager	URL not saved in Preferences; restart the IDE
"Teensy Loader is unable to..." (Windows)	you have the Microsoft Store IDE -> use the official installer
RGB_ORDER / template errors in FastLED	FastLED isn't 3.9.10 (downgrade to 3.9.10)
AudioPlaySdResmp / playRaw / override errors	the newdigate libraries are the registry ones -> install them from GitHub
nullptr / ResamplingReader errors	the patch from ch. 7 wasn't applied
'usbMIDI' was not declared	USB Type not set to Serial + MIDIx16
macOS: "operation not permitted" on the libraries	re-grant access to Documents in Privacy & Security
Port not visible	charge-only USB cable -> use a data cable

Part of the [ichosynth](#) project · fork of NI404 (SP_) · for the ICHOS 2026 workshop.